Histories as Events: the Order of the Pair Rule, First-Order Weights and Forces from Gains of Bounded Order

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Abstract

On a gain digraph, a family of set functions on walks of each length is one function on sets of unending walks exactly when the measure of a set of walks equals, up to a fixed factor, the measure of the set of all its continuations. Under that condition an additive rule is a multiple of the number of walks, so the weighting of returning walks by their holonomy that invariance under switching allows is excluded. On the signed arc digraph of a line, a rule that adds over end vertices a function of the modulus of the summed gains satisfies the condition only if the function is the square; on any norm-preserving gain digraph in which two walks merge, the only power of the modulus that satisfies it is the second. The second order of the pair rule of paper G8 thus follows from the condition, the form of the rule and the carrier. A position-dependent gain translates the quasi-momentum exactly only if it is linear in position, so a translation by Φ per step needs gains of order at least $2\pi/\Phi$. For gains of order three or more no quasi-energy lies in the spectrum of the free step multiplied by every gain value, apart from four band edges for order four; in computed instances a staircase of gains of order three, four or six confines a walk to a block, leaks slowly or is a strong field, and does not follow the ray of its mean field. For a linear gain the centroid of a packet follows the ray averaged over the packet, displaced by the connection of the branch eigenvector, with a remainder that falls about as the first power of the field in the instances computed. The statements are conditional on the event structure, the form of the rule and the carrier. Neither quantum nor classical mechanics is derived, and no physical correspondence is asserted.

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1 Question and results

Paper G8 assigns a measure to sets of walks of one length on a gain graph, and leaves three items as assumptions: the second order of the rule, the exclusion of a first-order weighting of returning walks, and any relation between the measures at different lengths [4]. Paper G9 takes the simplest such relation, consistency under continuation, as a requirement and finds what it forces on the carrier [5]. It leaves open a reason for the requirement, an error bound for the position of a ray, and whether gains of bounded order can imitate a weak force. The present paper asks what the same requirement decides about the rule itself, and returns to the two open items.

(i) A family of set functions on walks of each length defines one function on sets of unending walks exactly when it is consistent under continuation (Lemma 3.1). The requirement of G9 is thus the statement that events are sets of histories.

(ii) An additive rule that is consistent under continuation is a multiple of the number of walks: the weighting of returning walks by their holonomy, which switching invariance allows, is excluded (Proposition 4.1).

(iii) On the signed arc digraph of a line, a rule that adds over end vertices a function of the modulus of the summed gains is consistent under continuation only if that function is the square (Theorem 5.1). On any norm-preserving gain digraph with a merging pair of walks, the only consistent power of the modulus is the second (Theorem 5.2).

(iv) A gain that depends on position translates the quasi-momentum by a fixed amount per step exactly only if it is the linear gain; a translation by Φ therefore needs a gain of order at least $2\pi/\Phi$ (Proposition 6.1). For gains of order $q \geq 3$ no quasi-energy belongs to the spectra of the free step multiplied by all q gain values, except four band edges for $q = 4$ (Proposition 6.2). Computed instances of staircases of gains of order 3, 4 and 6 are recorded in Section 6; none follows the ray of its mean field.

(v) For the linear gain, the centroid of a packet follows the ray averaged over the quasi-momentum distribution of the packet, displaced by the connection of the branch eigenvector, with a remainder that decreases about as the first power of the field in the instances computed (Section 7). No bound is proved.

Section 8 states what is assumed and what is not obtained. The statements are mathematical and concern the constructed trials GH-1 and GH-2 of the program collection. Neither quantum nor classical mechanics is derived.

2 Setting

The notation is that of [5]. A gain digraph D has a finite or countable vertex set V, arcs e with initial vertex $s(e)$, terminal vertex $t(e)$ and gain $\varphi(e)$ in a group G, at most one arc from a given vertex to a given vertex, and finitely many arcs leaving each vertex; a character $\chi\colon G \to \mathrm{U}(1)$ is fixed and $A_{yx} = \chi(\varphi(e))$ for an arc e from x to y, $A_{yx} = 0$ if there is none. An event S is a set of walks of one length j from one vertex x, with vector $v_S(y) = \sum_{w \in S,\, t(w)=y} \chi(\varphi(w))$, and $\operatorname{Ext} S$ is the event of all walks of length $j+1$ whose first j arcs form a walk of S; $v_{\operatorname{Ext} S} = A v_S$. A family $(\mu_j)_{j \geq 0}$ of real functions on the events of each length is *consistent under continuation with factor* $c > 0$ if

$$\mu_{j+1}(\operatorname{Ext} S) = c\, \mu_j(S) \qquad \text{for every event } S \text{ of every length } j. \tag{2.1}$$

A *rule* gives μ_j by one formula for all j. The rule of [4] is $\mu(S) = \|v_S\|^2$; by Lemma 3.1 of [5] it satisfies (2.1) for all events exactly when $A^*A = c$ on the vectors concerned, and D is then called norm-preserving of degree c.

3 Events as sets of histories

Suppose every vertex has an outgoing arc. A *history* from x is an unending walk $(e_1, e_2, \dots)$ with $s(e_1) = x$. For a walk w of length j from x let $\mathrm{cyl}(w)$ be the set of histories whose first j arcs are w, and for an event S let $\mathrm{cyl}(S) = \bigcup_{w\in S} \mathrm{cyl}(w)$. The sets $\mathrm{cyl}(S)$ form an algebra of sets, the algebra of cylinder sets.

Lemma 3.1. *(a)* $\mathrm{cyl}(\mathrm{Ext}\, S) = \mathrm{cyl}(S)$*, and for events* S, S' *of lengths* $j \le j'$ *one has* $\mathrm{cyl}(S) = \mathrm{cyl}(S')$ *if and only if* $S' = \mathrm{Ext}^{j'-j} S$*.*

(b) A family (ν_j) *of functions on events defines a function* ν *on the algebra of cylinder sets by* $\nu(\mathrm{cyl}\, S) = \nu_j(S)$ *if and only if* $\nu_{j+1}(\mathrm{Ext}\, S) = \nu_j(S)$ *for all* S *and* j*.*

(c) A family (μ_j) *is consistent under continuation with factor* c *if and only if* $\nu_j = c^{-j}\mu_j$ *defines a function on the algebra of cylinder sets.*

Proof. (a) A history begins with a walk of S exactly when its first $j+1$ arcs form a walk of $\mathrm{Ext}\, S$. If $\mathrm{cyl}(S) = \mathrm{cyl}(S')$, a walk w' of length j' lies in S' exactly when the histories beginning with w' lie in $\mathrm{cyl}(S)$; since every walk is the beginning of a history, this holds exactly when the first j arcs of w' form a walk of S. (b) follows from (a), and (c) from (b). □

Condition (2.1) is therefore not a dynamical law added to the rule. It says that an event is a set of histories, and that the measure of a set of histories does not depend on the length at which the set is described. For a system of finitely many states with a unitary step the decoherence functional of Dowker, Johnston and Surya is defined on exactly this algebra, with cylinder sets of finite strings as generators [3]; they show that it does not extend to the generated σ-algebra in general, which is not used here. That events are sets of histories is an assumption about the event structure. It is supplied. What follows is what it decides once it is made.

4 Additive rules

By Lemma 3.1 of [4] a switching-invariant additive rule has the form $\mu(S) = \sum_{w\in S} F(w)$ with $F(w) = a$ for a walk that does not end at its starting vertex and $F(w) = F_{\mathrm{c}}(\varphi(w))$, a class function of the gain, for a walk that does. Such a rule is consistent under continuation with factor c exactly when

$$\sum_{e\colon s(e)=t(w)} F(we) = c\, F(w) \qquad \text{for every walk } w \text{ from } x. \tag{4.1}$$

Proposition 4.1. *Let* D *be a gain digraph without loops in which every vertex has* $d \ge 1$ *outgoing arcs, and let an additive rule of the form above satisfy* (4.1) *with a factor* $c > 0$*. If* $a \ne 0$*, then* $c = d$ *and* $F_{\mathrm{c}}(g) = a$ *for every* g *that is the gain of a closed walk at* x*. If* $a = 0$*, then* $F_{\mathrm{c}}(g) = 0$ *for every such* g*. In both cases* μ *is* a *times the number of walks.*

Proof. A closed walk w at x with gain g has d continuations, none of them closed, since there are no loops; so (4.1) reads $da = cF_{\mathrm{c}}(g)$. If $a = 0$ this gives $F_{\mathrm{c}}(g) = 0$. Let $a \neq 0$. If some walk from x ends at a vertex $y \neq x$ without an arc to x, its continuations are open and (4.1) reads $da = ca$, so $c = d$ and then $F_{\mathrm{c}}(g) = a$. Otherwise every walk w ending at $y \neq x$ has exactly one closed continuation, with some gain g', and (4.1) reads $(d-1)a + F_{\mathrm{c}}(g') = ca$. Every closed walk of positive length arises in this way, so F_{c} has the single value $f = (c-d+1)a$ on their gains; with $da = cf$ this gives $(c-d)(c+1) = 0$, hence $c = d$ and $f = a$. The closed walk of length 0 gives $da = cF_{\mathrm{c}}(1)$ and the same value. □

The weights 2 and $1/2$ of the example of Directors A, B and G for $\mathbb{Z}_3$, quoted after Lemma 3.1 of [4], are thus not the values of a function on sets of histories: on the complete graph on four vertices with one gain of order three the ratios in (4.1) take the values 1.5, 2.5 and 4 on walks of length two [2]. For two triangles with a common vertex, where the out-degrees are 2 and 4, no factor among those tried makes a rule with $a \neq 0$ consistent [2]. A first-order rule on histories is the count of walks, which is the uniform classical measure on histories and does not depend on the gains.

5 Rules that depend on the modulus

Let $\phi\colon [0,\infty) \to \mathbb{R}$ with $\phi(0) = 0$, and put

$$\mu_\phi(S) = \sum_{y \in V} \phi(|v_S(y)|). \tag{5.1}$$

This keeps two assumptions of [4]: gains of walks with a common end vertex are added, and end vertices are records over which the rule adds. It drops the second order. The rule of [4] is $\phi(t) = t^2$, its rule of fourth order is $\phi(t) = t^4$, and with trivial gains $\phi(t) = t$ is the number of walks.

5.1 The signed line

Let D be the signed arc digraph of the line $\mathbb{Z}$ of [5]: vertices (x, R) and (x, L), arcs from either to $(x+1, R)$ and to $(x-1, L)$, all gains $+1$ except -1 on the arcs from (x, L) to $(x-1, L)$. The vectors v_S have integer components, so only the values of ϕ at integers enter.

Theorem 5.1. *On the signed arc digraph of the line, the rule* (5.1) *with* $\phi(1) \neq 0$ *is consistent under continuation with a factor* $c > 0$ *if and only if* $c = 2$ *and* $\phi(n) = n^2\phi(1)$ *for every integer* $n \geq 0$*.*

Proof. For one walk w, $\mu_\phi(\{w\}) = \phi(1)$ and $\mathrm{Ext}\{w\}$ consists of two walks with different end vertices, so $2\phi(1) = c\,\phi(1)$ and $c = 2$. Let $a \geq 1$. A walk from $(0, R)$ is a sequence of steps to the right and to the left, and its gain is -1 to the power of the number of steps to the left that follow a step to the left. Among the walks of length $a+1$ with a steps to the right and one to the left there are a that end with a step to the right and one that ends with the step to the left; all have gain $+1$ and end at site $a-1$. Let S consist of these $a+1$ walks. Then v_S has the components a at $(a-1, R)$ and 1 at $(a-1, L)$ and no others, and $v_{\mathrm{Ext}\,S} = Av_S$ has the components $a+1$ at (a, R) and $a-1$ at $(a-2, L)$. Consistency gives

$$\phi(a+1) + \phi(a-1) = 2\phi(a) + 2\phi(1), \qquad a \geq 1,$$

and with $\phi(0) = 0$ induction gives $\phi(n) = n^2\phi(1)$. Conversely $A^*A = 2$ and $\phi(t) = t^2$ is consistent by Lemma 3.1 of [5]. □

The recursion is the parallelogram identity $\phi(|a+b|)+\phi(|a-b|) = 2\phi(|a|)+2\phi(|b|)$ at $b = 1$; the full identity holds at every site because the two arcs arriving at a site continue to one vertex with equal signs and to another with opposite signs, which is the zigzag gain -1 of Theorem 4.3 of [5]. A numerical solution of the consistency equations over 4000 sampled events of length 12 returns $\phi(n)/\phi(1) = n^2$ for $n \le 8$ to 3×10^{-13}, and rejects $\phi(n) = n$ and $\phi(n) = n^4$ [2]. With all gains $+1$ on the same digraph the rule $\phi(t) = t$ is consistent, being the count.

5.2 Norm-preserving digraphs

Theorem 5.2. *Let D be a gain digraph with gains of modulus one and $A^*A = d$, $d \ge 2$. Suppose two walks of equal length from a common vertex end at distinct vertices y_1, y_2 that have a common successor. Then for $p > 0$ the rule $\mu_p(S) = \sum_y |v_S(y)|^p$ is consistent under continuation with some factor $c > 0$ if and only if $p = 2$.*

Proof. Single walks give $c = d$, as every vertex has d successors. Let w_1, w_2 be the two walks, a, b the values of χ on their gains, and $S = \{w_1, w_2\}$, so $\mu_p(S) = 2$. Let Z be the set of the $m \ge 1$ common successors of y_1, y_2. For $z \in Z$ the component of Av_S is $aA_{zy_1} + bA_{zy_2}$, of modulus $|1 + u_z|$ with $u_z = (b/a)\,\overline{A_{zy_1}}A_{zy_2}$; the other $2(d - m)$ successors carry components of modulus one. Orthogonality of the columns y_1, y_2 of A gives $\sum_{z\in Z} u_z = 0$. Hence $\mu_p(\operatorname{Ext} S) = 2(d - m) + h(p)$ with $h(p) = \sum_{z\in Z} |1 + u_z|^p$, and consistency requires $h(p) = 2m$. Now $h(2) = \sum_z (2+2\operatorname{Re} u_z) = 2m$. On $(0, \infty)$ the function h is a sum of functions t^p with $t \ge 0$ and is convex, and its limit at 0 is the number $h_0 \le m$ of z with $u_z \ne -1$. For $0 < p < 2$ convexity gives $h(p) \le (1-p/2)h_0 + (p/2)\,2m < 2m$, and for $p > 2$ it gives $h(p) \ge 2m + (p-2)(2m - h_0)/2 > 2m$. □

The hypothesis on the pair of walks is the hypothesis of Lemma 3.1(b) of [5] together with branching; it holds on the ring and the line. Theorem 5.2 is the statement, for the vectors of events, that a unitary matrix which is not a permutation matrix with phases preserves no ℓ_p norm other than the Euclidean one; for all vectors this is the classical description of the isometries of ℓ_p, $p \ne 2$, as permutation matrices with phases, attributed to Li and So in the finite-dimensional case [7]. That reference was not inspected for this paper and is not used in the proof.

Taken together with Proposition 4.1 this gives the following division, under the assumption that events are sets of histories. A rule of first order is the count and is blind to the gains. A rule of the form (5.1) that depends on the gains and is consistent on a branching norm-preserving carrier is the squared modulus. The second order of the rule of [4] is then a consequence and no longer a separate assumption. The form (5.1), the record role of end vertices and the carrier remain assumed, and rules that are not sums over end vertices of a function of one modulus are not covered.

6 Gains of bounded order and a force

Let U be the normalized step of the signed arc digraph of the line or of a ring, $\mathcal{T}$ the translation by one site, and for a function m of modulus one on the sites let M_m be the multiplication by $m(x)$; UM_m is the step in which every arc leaving site x carries

the additional gain $m(x)$. Proposition 6.3 of [5] shows that $m(x) = e^{-i\Phi x}$ translates the quasi-momentum by $-\Phi$ per step.

Proposition 6.1. *$\mathcal{T}\,UM_m\,\mathcal{T}^{-1} = \lambda UM_m$ for a number λ if and only if $m(x-1) = \lambda\, m(x)$ for all x, that is, $m(x) = m(0)\lambda^{-x}$. If m takes its values in a finite group of roots of unity, λ belongs to that group; an exact translation of the quasi-momentum by $\Phi = 2\pi/n$ per step requires gains of order at least n.*

Proof. $\mathcal{T}$ commutes with U and $\mathcal{T}M_m\mathcal{T}^{-1}$ is the multiplication by $m(x-1)$; U is invertible. □

With gains of bounded order an exact force is therefore the linear gain of that order, a strong field. A weak mean field can only be approximate, with a gain that winds slowly: constant on long blocks and advancing by one group element from block to block, as in the staircase of [5]. On a block where the gain is the constant γ the step is γU. By Lemma 6.1 of [5] the spectrum of U on the line is

$$\Sigma = \{e^{i\theta} : |\theta| \le \pi/4\} \cup \{e^{i\theta} : |\theta - \pi| \le \pi/4\},$$

two arcs of length $\pi/2$ separated by two gaps of length $\pi/2$.

Proposition 6.2 (Local spectra)**.** *Let $\zeta = e^{2\pi i/q}$. The set $\bigcap_{j=0}^{q-1} \zeta^j \Sigma$ equals Σ for $q = 1, 2$, consists of the four points $e^{\pm i\pi/4}, e^{\pm 3i\pi/4}$ for $q = 4$, and is empty for $q = 3$ and for $q \ge 5$.*

Proof. A point lies in the intersection exactly when its orbit under multiplication by ζ avoids both open gaps. $\Sigma = -\Sigma$ settles $q \le 2$. For $q \ge 5$ consecutive points of an orbit are at angular distance $2\pi/q < \pi/2$, and an open arc of length $\pi/2$ contains one of them. For $q = 4$ the distance is $\pi/2$ and an orbit avoids a gap only by containing both its end points. For $q = 3$ two of the three points of an orbit would lie in one arc of length $\pi/2$, at distance $2\pi/3$. □

For $q \ge 3$ no quasi-energy is thus an interior point of the spectrum on every block of a staircase that winds through the group; for $q = 4$ the common points are band edges, where the group velocity vanishes. A state that propagates in one block is in a gap of another. This is a statement about the spectra of the pieces. It is not a theorem about the dynamics of the staircase, whose gains are periodic in position, so that the theorem of Grimmett, Janson and Scudo [6] applies with the period as cell and exact confinement is not to be expected. The computed instances are as follows [2].

block length L	1	2	4	16	64
order 2	1108	750	403	346	347
order 3	520	520	160	9.3	31
order 4	750	365	102	8.4	34
order 6	520	181	27	11	48

Table 1: Root-mean-square distance, in sites, after 2048 steps of a walk from one vertex under a staircase of gains of order q with blocks of L sites; $L = 1$ is the linear gain of order q. Ring of 8192 sites; the walk does not reach the far side [2].

Observed. (a) For $L = 1$ the distance doubles between 1024 and 2048 steps: a linear gain with a rational field spreads ballistically, as proved by Cedzich et al. [1]; the bounded distance reported in Section 6.2 of [5] refers to four periods. (b) For $L = 16$ and $L = 64$ and the orders $3, 4, 6$ the walk stays within about one block up to 2048 steps; for $L = 4$ it does not. Order 2 spreads for every L, in accordance with $\Sigma = -\Sigma$. (c)

A packet meeting one interface with gain ratio $e^{-i\delta}$ is transmitted with probability 0.9997, 0.9994, 0.975 for $\delta = \pi/12, \pi/6, \pi/4$, with probability 0.0025 for $\pi/3$, below 10^{-12} for $\pi/2$ and $2\pi/3$, and 0.48 for π; the packet has quasi-energy near -0.21, which lies in the shifted band for $\delta < 0.996$. (d) Under a staircase of order 4 a packet is displaced, after an eighth of the period of the mean field, by 0.69, 0.54 and 0.16 of its displacement under the linear gain of the same mean slope, for $L = 16, 64, 256$: the deviation from the mean field grows with the block length. The first two values reproduce those of [5].

Failed expectations, kept. Two predicates fixed before their runs failed. The first expected the staircase of period 64 to drift within 1024 steps; it stayed within four sites of its start. The second expected confinement for $L = 4$. A computation of band velocities from sorted eigenphases returned values above the largest possible speed and below the observed spread, and is discarded as an instrument. All three are in the result files.

In these instances a winding gain of order 3, 4 or 6 either is a strong field, or holds the walk within a block, or leaks slowly; in none does a packet follow the ray of the mean field. Whether some other arrangement of gains of bounded order has a ray limit with a force is not decided.

7 The position of a ray under a linear gain

Table 4 of [5] compares the centroid of a packet with the ray $\dot{k} = -\Phi$, $\dot{x} = v(k)$ through the central quasi-momentum k_0, and reports a deviation of about 0.6 site, most of it the change of the connection $\mathcal{A}(k) = \mathrm{Im}\langle u(k)|\partial_k u(k)\rangle$ of the branch eigenvector. Extending that comparison to longer periods shows that the remainder does not decrease: it is 0.075, 0.038, 0.038, 0.062, 0.117, 0.232 site at the periods 64 to 2048, and a predicate that expected a first-order decrease failed [2]. The cause is the width of the packet in k. If the ray and the connection are averaged over the quasi-momentum distribution ρ of the initial packet,

$$\bar{x}(t) = \sum_{s=1}^{t} \int v\big(k - (s - \tfrac{1}{2})\Phi\big)\, \rho(k)\, dk - \int \big[\mathcal{A}(k - t\Phi) - \mathcal{A}(k)\big]\rho(k)\, dk, \tag{7.1}$$

with the signs of the conventions of [5], the remainder decreases (Table 2); the fitted exponent against the period is -1.1.

period $2\pi/\Phi$ (steps)	64	128	256	512	1024	2048
ray through k_0, with connection	0.0748	0.0385	0.0381	0.0622	0.1174	0.2315
averaged ray (7.1)	0.0735	0.0273	0.0135	0.0071	0.0029	0.0016

Table 2: Largest deviation, in sites, of the centroid from the prediction over one period. Packet at $k_0 = 1$ on the first branch, width 0.03 in k, ring of 8192 sites. The second row is sampled at 32 times per period. For a packet of width 0.015 the second row gives 0.0075 and 0.0018 at the periods 512 and 2048 [2].

Equation (7.1) is what the adiabatic approximation gives: a vector that stays on one branch has the form $c(k + t\Phi)\, e^{-i\int\omega} u(k)$ in the quasi-momentum representation, in which position is $i\partial_k$, so that the mean position is the mean derivative of the phase plus the mean of $\mathcal{A}$. For an electron in a periodic potential with slowly varying external potentials Panati, Spohn and Teufel prove that the dynamics in an isolated

band is governed, to first order in the slowness parameter, by semiclassical equations in which the Berry connection of the band enters [8]. Their theorem concerns Schrödinger operators in continuous time. No corresponding bound is proved here for the step of this paper; the criterion of 0.02 site of [5] is met by the averaged prediction from the period 256 on, and the failed predicate of [5] stands for the ray through k_0.

8 Assumptions and limits

Assumed. (1) Events are sets of histories (Section 3). This replaces "consistency under continuation" in the list of [5] by a statement about events; it is not derived from the carrier. (2) The form (5.1): summed gains at a common end vertex, a function of their modulus, addition over end vertices. (3) The carrier: for Theorem 5.1 the signed line, for Theorem 5.2 a norm-preserving digraph with a merging pair of walks. (4) The group and the character, as in [4].

Obtained under these assumptions. The second order of the rule; the exclusion of first-order weights of returning walks.

Not obtained. (5) A reason for (1) to (4). In particular nothing here selects a carrier on which a rule that depends on the gains is consistent: on a digraph of constant out-degree with trivial gains the count is consistent and classical. (6) A classification of consistent rules outside the form (5.1). (7) A ray limit with a force for gains of bounded order, or a proof that there is none; Proposition 6.2 concerns local spectra only. (8) An error bound for (7.1). (9) Everything listed as not obtained in [4, 5]: a clock, a selection of complex amplitudes, composition compatible with the step, and any physical correspondence.

9 Status of the statements

Statement	Kind
Lemma 3.1; Propositions 4.1, 6.1, 6.2; Theorems 5.1, 5.2	derived results, proofs given here
Tables 1, 2; the values in Sections 4, 5, 6, 7	computed instances
Equation (7.1)	heuristic; compared with computed instances; not proved

The components are elementary or known: cylinder sets of histories, the parallelogram identity as a characterization of the square, the isometries of ℓ_p, band spectra of a walk with a piecewise constant phase, and the semiclassical motion in a band with its Berry connection. The paper applies them to the pair measure on a gain digraph and claims no priority. The statements have not been refereed, and the numerical tables are not error enclosures.

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